

# T3 Weinberg-coefficient RGE report

$T3 - D$ ,  $\alpha = -2$ :  $S_1 = (3, -1)$ ,  $S_2 = (1, 0)$ ,  $F = (2, -\frac{1}{2})$

|                        |               |                           |         |
|------------------------|---------------|---------------------------|---------|
| <b>Matched EFT RGE</b> | <b>Failed</b> | <b>Flavor RGE</b>         | Not run |
| <b>Neutrino mass</b>   | Not run       | <b>Numerical running</b>  | Not run |
| <b>Observables</b>     | Not run       | <b>Internal benchmark</b> | Not run |

## 1 Conventions and scope

Below the common heavy-particle threshold, the active theory is SMEFT.

$$t = \ln \mu,$$

$$V(H) = \frac{\lambda_H}{2} (H^\dagger H)^2.$$

The flavor construction assumes three lepton generations, a diagonal heavy-fermion mass basis, and common scalar masses.

## 2 Matched Weinberg coefficient

$$C_5 = -\frac{2}{3} \left( -\frac{\sqrt{3}hM_F^3M_{S_2}^2\lambda_{T3}\log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} + \frac{\sqrt{3}hM_F^3M_{S_1}^2\lambda_{T3}\log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} - \frac{\sqrt{3}hM_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_1}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} + \frac{\sqrt{3}hM_F M_{S_1}^2 M_{S_2}^2 \lambda_{T3} \log\left(\frac{M_F^2}{M_{S_2}^2}\right)\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}}{(M_F^2-M_{S_1}^2)(M_F^2-M_{S_2}^2)(M_{S_1}^2-M_{S_2}^2)} \right)$$

## 3 One-generation SMEFT check

The tensor calculation is required to reproduce the compact one-generation benchmark

$$16\pi^2\beta_{C_5} = C_5 \left[ -3g_2^2 + 2\lambda_H + 6|y_u|^2 + 6|y_d|^2 - |y_e|^2 \right]. \quad (1)$$

The RGE stage stops with an error if the calculated expression does not reduce to this benchmark.

## 4 Three-generation flavor result

The matched one-generation result is factorized as

$$C_5^{(1)} = F_{\text{loop}} \bar{y}_1 \bar{y}_2,$$

and lifted to the symmetric flavor matrix

$$(C_5)_{pq} = \frac{1}{2} \sum_r F_r \left[ y_{1,pr}^* y_{2,qr}^* + y_{2,pr}^* y_{1,qr}^* \right]. \quad (2)$$

Writing this matrix as  $K \equiv C_5$ , the implemented RGE is

$$16\pi^2 \frac{dK}{d\ln\mu} = (2\lambda_H - 3g_2^2 + 2T)K - \frac{3}{2} [Y_e Y_e^\dagger K + K(Y_e Y_e^\dagger)^T], \quad (3)$$

$$T = \text{Tr} \left( Y_e Y_e^\dagger + 3Y_u Y_u^\dagger + 3Y_d Y_d^\dagger \right). \quad (4)$$

The fully expanded symbolic matrix is retained as a data file rather than printed here, because it is much less readable than the matrix equation above.

## 5 Neutrino-mass conversion

The current symbolic and numerical implementation uses

$$m_\nu = -v^2 C_5. \quad (5)$$

**Normalization check required.** Earlier function documentation stated  $m_\nu = -(v^2/2)C_5$ , whereas the executed code uses  $m_\nu = -v^2 C_5$ . The extracted operator normalization and this conversion must be checked together before final phenomenology.

## 6 Numerical running and observables

*Numerical running was not requested. Supply `-numerical CONFIG.json` to include this section.*

## 7 Saved data

| Result                       | File                    |
|------------------------------|-------------------------|
| Matched coefficient          | data/c5_coefficient.txt |
| Rendered matched coefficient | c5_coefficient.pdf      |